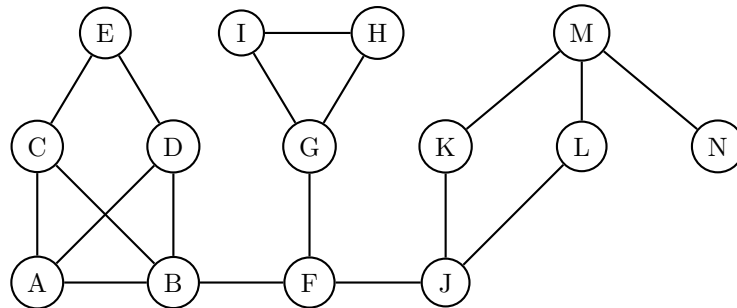


Final examination

Documents are not allowed. Electronic devices are not allowed.

Exercise 1 — *Simple questions on a simple graph (5 points)*

Consider the following graph.



- Q1.** Draw the degree distribution of the graph.
- Q2.** Give a possible BFS and DFS ordering of the nodes starting from node G.
- Q3.** What is the diameter of the graph? Justify briefly.
- Q4.** Compute the core value of each node.
The betweenness centrality of a node is the number of shortest paths passing by the node.
- Q5.** Which node has the highest betweenness centrality? Justify briefly.
Compute the betweenness centrality of nodes B, F and M.

Exercise 2 — *Real-world graph algorithmics (5 points)*

- Q1.** Describe an efficient algorithm to list all triangles in the input graph. Explain briefly why your algorithm is efficient on large sparse real-world graphs.
- Q2.** Describe an algorithm that computes a lower bound to the diameter of the input graph. The lower bound should be empirically good for large sparse real-world graphs and the algorithm should run in linear time. Give some intuition of why the lower bound is empirically good.
- Q3.** Describe the PageRank algorithm based on the power iteration method seen in class.

For each description, make sure that you detail all important steps. You can include some pseudocodes.

Exercise 3 — *Number of bits to store a graph (5 points)*

Consider a graph with n nodes and m edges. The graph is unweighted, directed and can contain self-loops.

Assume that you need 8 bits to store a boolean and 64 bits to store an integer.

- Q1.** How many bits do you need to store the graph as an adjacency matrix? Justify briefly.
- Q2.** How many bits do you need to store the graph as a list of edges? Justify briefly.
- Q3.** Assume the density of the graph $d = \frac{m}{n^2}$ is equal to 0.1. Which one of the two formats requires less bits? Same question with $d = 0.001$.
- Q4.** How many bits do you need to store the graph as an adjacency array. Justify briefly.

Now assume that you need 1 bit to store a boolean and $\lceil \log_2(n) \rceil$ bits to store an integer such that the maximum integer is n .

- Q5.** How many bits do you need to store the graph as an adjacency matrix? Justify briefly.
- Q6.** How many bits do you need to store the graph as a list of edges? Justify briefly.
- Q7.** Which of the two formats requires less bits as a function of the density d and the number of nodes n ?
- Q8.** How many bits do you need to store the graph as an adjacency array. Justify briefly.

Exercise 4 — *Extreme case (5 points)*

For any given $n \in \mathbb{N}$, let G_n be the graphs composed of $2n + 1$ nodes and $3n$ edges such that:

- a unique node n_0 is connected to all other nodes in the network;
- all other nodes have degree 2.

- Q1.** Draw the case G_4 .
 - The clustering coefficient of a node is defined as the number of triangles the node belongs to divided by its number of pairs of neighbors. The clustering coefficient of a graph is defined as the average clustering coefficient of all its nodes.
 - The transitivity ratio is defined as three times the number of triangles divided by the number of V-edges (a V-edge is a pair of edges that share an end node).
- Q2.** Compute the clustering coefficient of G_4 and the transitivity ratio of G_4 .
- Q3.** How do these coefficients evolve when n goes to ∞ ?
- Q4.** Provide an interpretation of these coefficients in terms of probability.
- Q5.** Think about the Facebook graph (users connected by friendship links). Which one of the clustering coefficient and the transitivity ratio do you think is the higher and why?